Placebo Trial Design Student Handout

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	**Time**	30-40 min
Difficulty	Medium	**ID**	placebo-trial-design

Big idea: Controls protect knowledge from bias.

Student-Facing Prompt

Inspect Placebo Trial Design Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Placebo Trial Design Stimulus A	Student-facing stimulus 1: A mock treatment trial with control, placebo, randomisation, and blinding cards. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Placebo Trial Design Stimulus B	Student-facing stimulus 2: A flawed trial students must repair before trusting its conclusion. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Placebo Trial Design Reveal / Twist	Student-facing stimulus 3: A debrief on ethics when belief itself changes outcomes. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Controls protect knowledge from bias.

Actual Lesson Questions

#	Question for students
1	After inspecting Placebo Trial Design Stimulus A, what is your first knowledge claim?
2	Which exact detail from Placebo Trial Design Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Why is personal experience not enough in science?

Ready-to-Use Examples

- A mock treatment trial with control, placebo, randomisation, and blinding cards.
- A flawed trial students must repair before trusting its conclusion.
- A debrief on ethics when belief itself changes outcomes.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Activity Steps

1. Groups design a fair test of the product.
2. They must include a control group, placebo, blinding, outcome measure, sample size and ethics note.
3. Groups exchange designs and find weaknesses.
4. The class agrees minimum standards for a trustworthy test.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Why is personal experience not enough in science?
- How do controls improve knowledge?
- Can scientific methods remove all bias?

Knowledge Questions

- Why is personal experience not enough in science?
- How do controls improve knowledge?
- Can scientific methods remove all bias?

Transfer Example

For example, run this activity with a mock treatment trial with control, placebo, randomisation, and blinding cards. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...